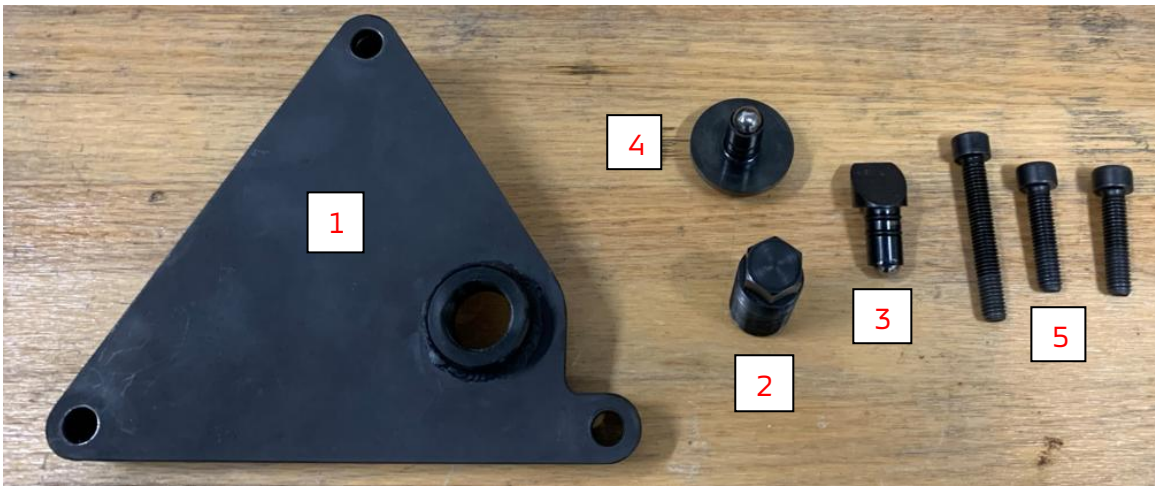
MMAL DEALER BULLETIN

Bulletin Subject	**Coolant seepage/leakage from core plug at rear of cylinder head on 4N15 Engine – Correction Type 2 Instructions**		
Applicable model(s)	MQ/MR Triton, QE/QF Pajero Sport		
Notice Category	**Hotline fix**	Technical Bulletin	**MQ/MR/QE/QF14-001**
Service Group	14 - Engine Cooling	Latest Revision Issued	15/12/2025
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Required Tools/Equipment

SST EG100141 is comprised of 5 main components – Refer G/L MMAL-25209.

1. Installer base plate.
2. Installer screw.
3. Core Plug tipping head.
4. Core Plug installation head.
5. Base plate retaining bolts.



Generic tools required to complete the repair –

1. General hand tools (wrench, sockets (external/internal hex), screw drivers, etc).
2. Small pry bar (approx 30cm in length).
3. Panel work stand.
4. Masking tape.
5. Lint free cleaning cloth.
6. Brake Cleaner.
7. Deburring tool/s (or similar).

Other Parts required – (refer main Hotline Fix document for details)

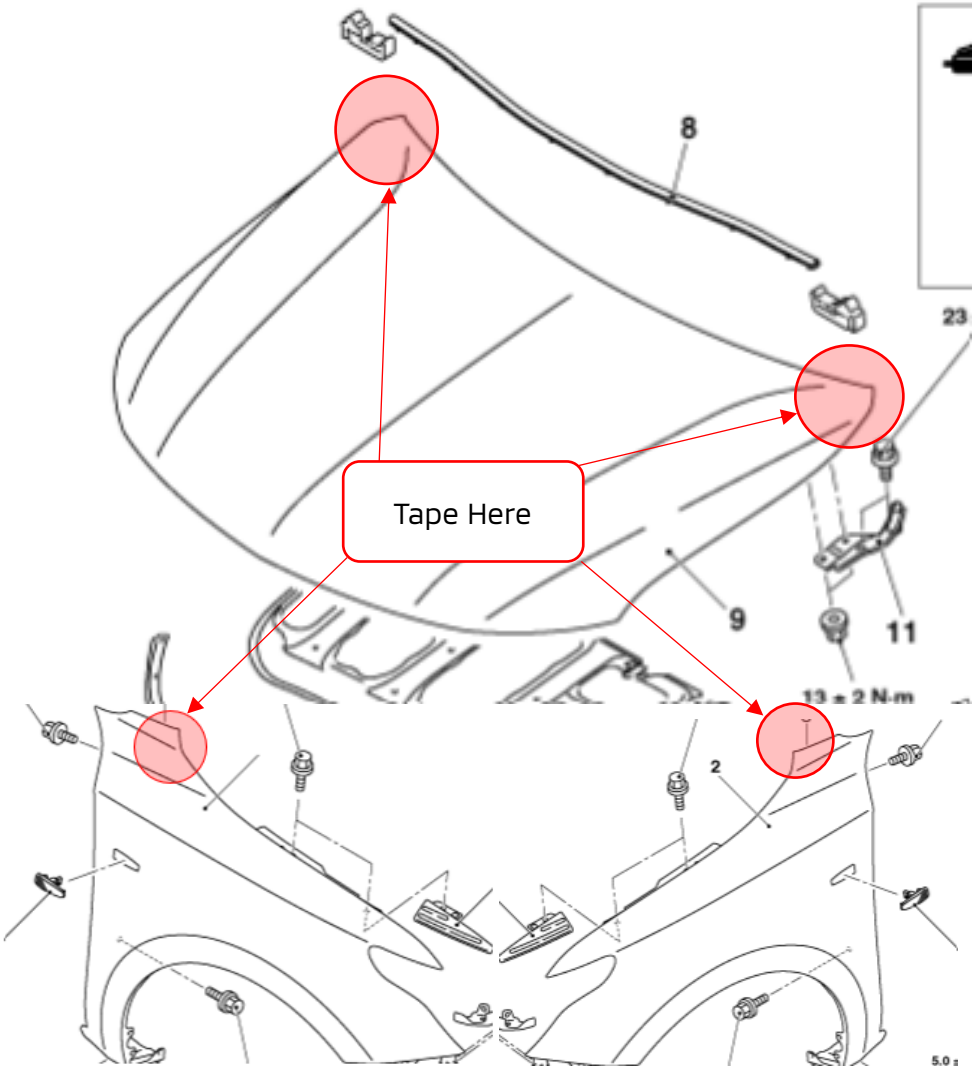
1. New Core Plug
2. Threebond 1360F adhesive

Repair Method

1. Drain the engine coolant as per WSM instructions.

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2. Inspect and take photos of any existing damage to the hood panel and fenders. Place masking tape on the upper corners of the front fenders and rear corners of the hood panel (4 locations, refer to images below). Also place tape around hood hinge locations (to mark current position), then remove the hood as per WSM instructions and place it on a panel work stand, in a safe location.



3. Remove the Front Deck Panel following the removal instructions in the WSM (including all pre removal steps shown below - 16MY Triton shown for example).
    - Front Deck Garnish Removal and Installation (Refer to GROUP 51, Front Deck Garnish).

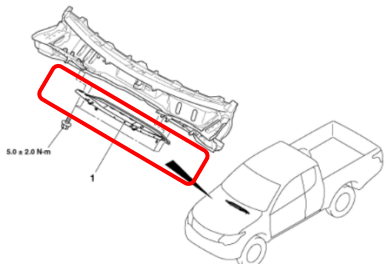
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- Windshield Wiper Link Assembly Removal and Installation (Refer to GROUP 51, Windshield Wiper .)
- Brake booster vacuum hose connection (Refer to GROUP 35A - Master Cylinder Assembly and Brake Booster Assembly <RHD>
- Brake booster vacuum pipe Removal and Installation (Refer to GROUP 35A - Master Cylinder Assembly and Brake Booster Assembly <RHD>

FRONT DECK PANEL REMOVAL AND INSTALLATION

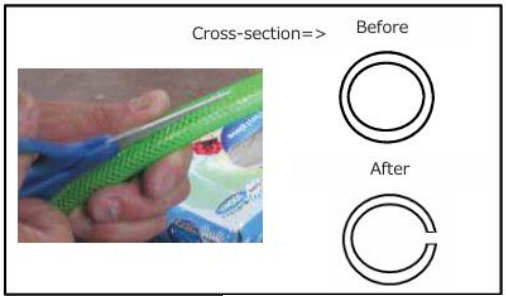
Pre-removal and Post-installation Operation

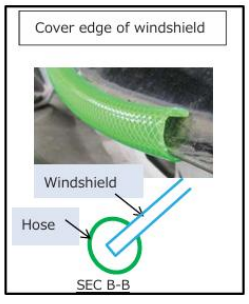
- Front Deck Garnish Removal and Installation (Refer to GROUP 51, Front Deck Garnish).
- Windshield Wiper Link Assembly Removal and Installation (Refer to GROUP 51, Windshield Wiper .)
- Brake booster vacuum hose connection (Refer to GROUP 35A - Master Cylinder Assembly and Brake Booster Assembly <LHD>, <RHD>
- Brake booster vacuum pipe Removal and Installation (Refer to GROUP 35A - Master Cylinder Assembly and Brake Booster Assembly <LHD>, <RHD>

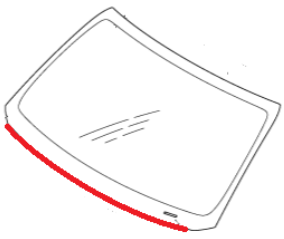


		Removal steps
	·	Engine control harness connection (Front deck panel side)
	·	Font harness connection (Front deck panel side)
<<A>>	1.	Front deck panel

Caution: Do not impact or place pressure on the edges of the front windscreen. Such actions may result in breakage of the glass. It is recommended that a slit section of hose is used to protect the lower edge of the windscreen glass across the area shown in red.

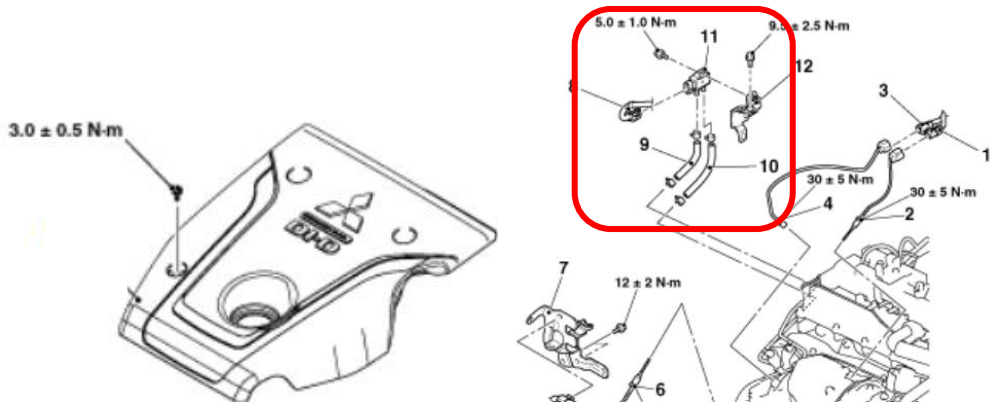




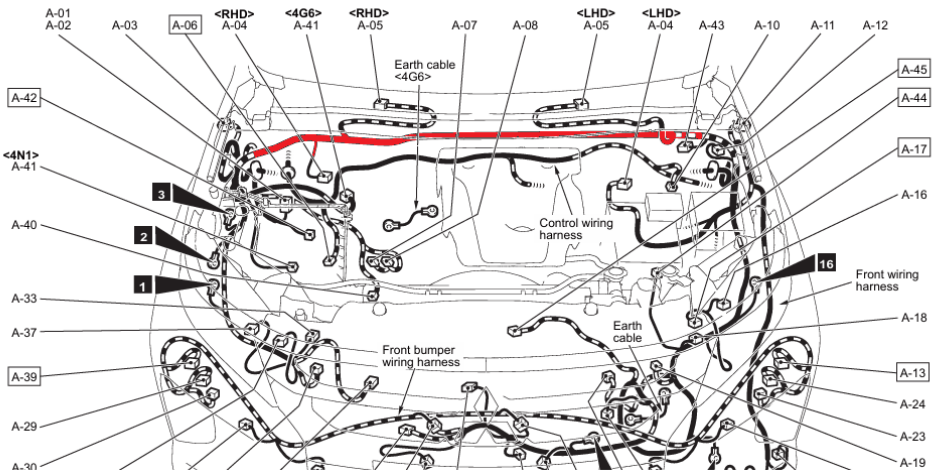


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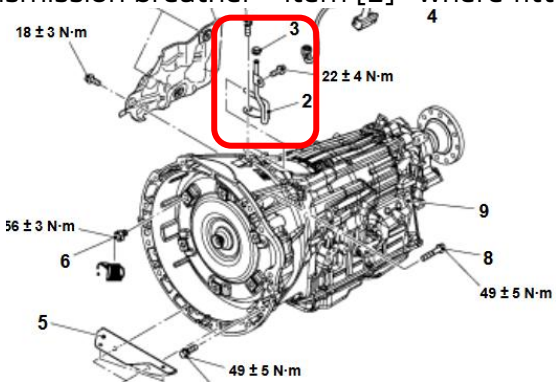
4. Remove the engine cover, differential pressure sensor and hoses as per the WSM.



5. Unclip the front wiring harness from the firewall (area shown in red) and move aside to allow better access to the rear of the cylinder head.

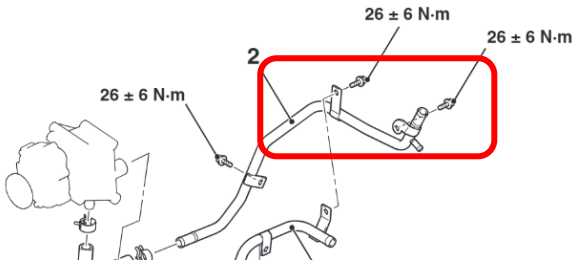


6. Remove the transmission breather - item [2] *where fitted

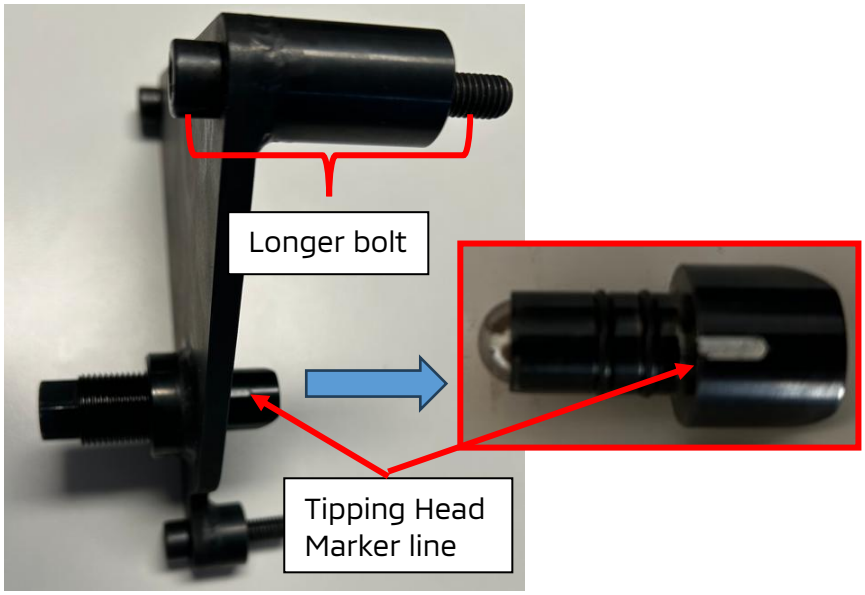


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7. Remove the 2 bolts retaining the rear coolant pipe, but do not remove the pipe.



8. Assemble the installer base plate [1], installer screw [2], tipping head [3] and base plate retaining bolts [5] as shown. The longer retaining bolt is located in the uppermost mounting hole.
Note: The tipping head must always be held at the 12 o'clock position (marker line facing directly upwards) during operation.

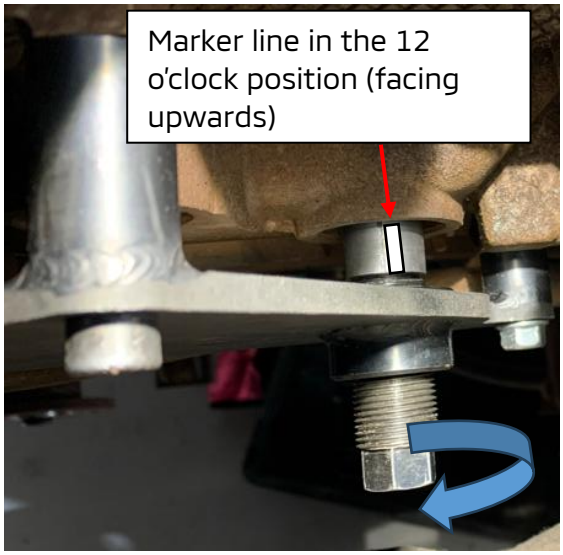


9. Install the tool to the three points of the cylinder head shown circled below. The base plate retaining bolts should be torqued to approximately 13 +/- 3Nm to ensure the base plate is sufficiently seated.
Note: As the upper threaded hole in the cylinder head is not normally utilised, the thread may need to be cleaned using an appropriate thread tap or similar prior to installation. The installer base plate is designed to be installed over the cooling pipe mounting tags, so do no move the cooling pipe to fit the installer base plate.

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10. Using an appropriate spanner or wrench, turn the installer screw clockwise, keeping the tipping head marker line in the 12 o'clock position until it firmly contacts the core plug.



11. Turn the installer screw (making sure the marker line remains at the 12 o'clock position) until the screw thread is no longer visible on the rearward side of the installer plate, or the core plug has released and tipped (the installer screw should be easier to turn once the core plug has fully released and tipped).

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12. Remove the tool assembly completely and position the core plug in a horizontal orientation using your finger* or a flat screwdriver.



Using the back side** of a pry bar facing rearwards, gently lever the core plug, taking care not to damage the flat surface surrounding the core plug bore and the bore itself.



The cap should wedge about ½ way out of the bore. Turn the pry bar around so the front side** of the pry bar is facing rearwards and lever the core plug out of the bore until it is fully removed, again taking care not to damage the flat surface surrounding the core plug bore and the bore itself.

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*__Caution:__ Sharp edges - Gloves are recommended when using your finger to position the core plug.
**Pry bar image below showing front and rear sides:



13. Remove any sharp edges from the core plug bore and rear machined surface using a deburring tool or similar, taking care not to not introduce contaminates to the cooling jacket or remove too much material from the bore surface (the adhesive sealant used in this process can seal up small gaps caused by minor scratches and imperfections, but will not be effective if the bore becomes too misshapen from heavy grinding, or use of extreme abrasives, etc).

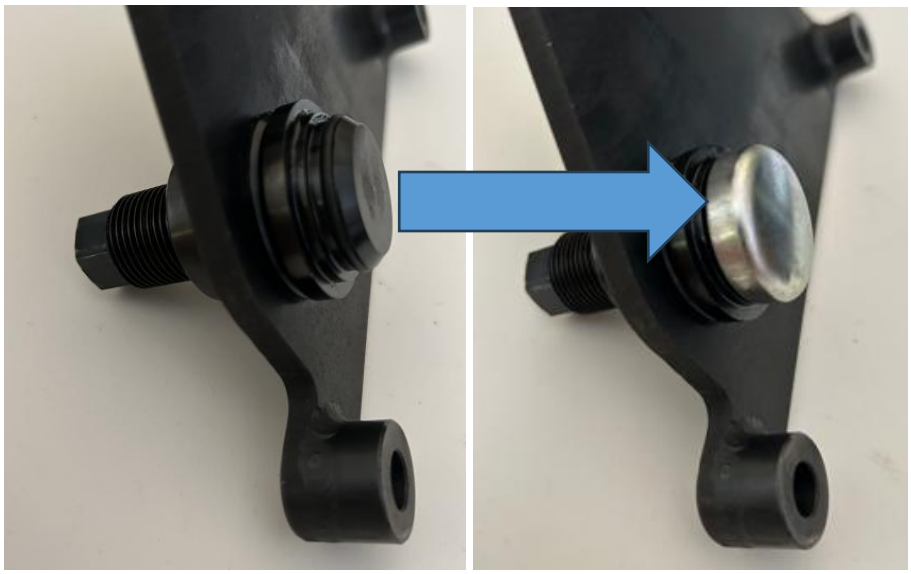
14. Clean the core plug bore and rear machined surface with a lint free cloth and brake cleaner (or similar fast evaporating solvent spray). A suction tool may be required to remove any standing coolant inwards of the core plug bore to ensure the bore is clean and dry before plug installation.



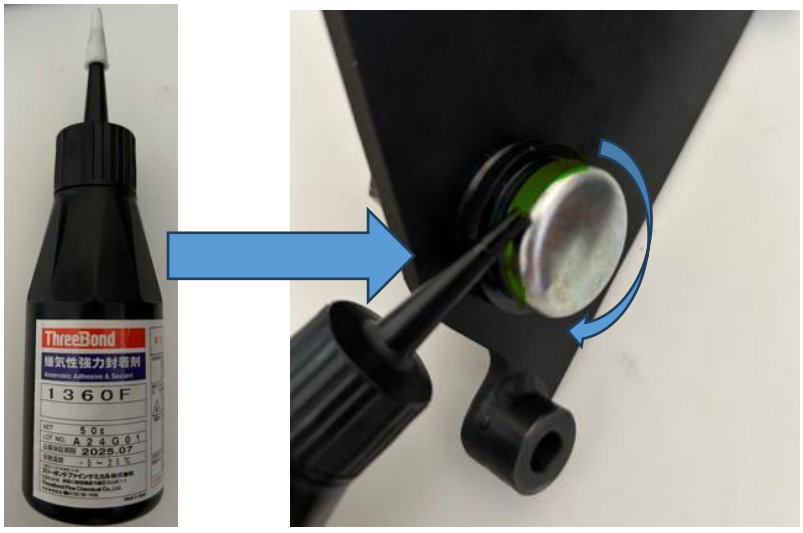
15. Remove the tipping head from the installer screw and fit the Core Plug installation head [4]. Wind the installer screw anti-clockwise from the installer base plate until the installation head is level with the forward

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facing side of the plate. Place the new core plug on the installation head so it is secured by the o-ring.



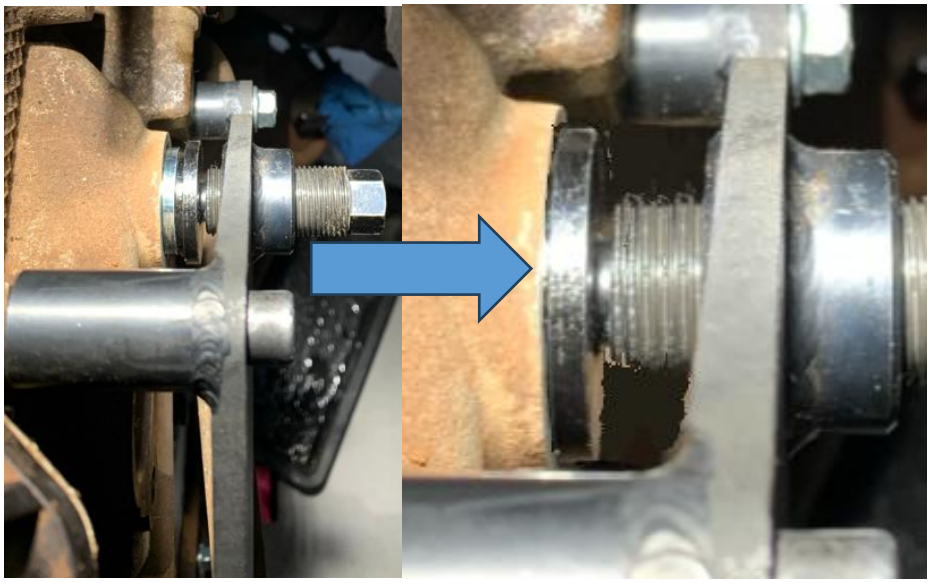
16. Using the Threebond 1360F adhesive, apply a small (approx 1mm) bead around the entire circumference of the core plug mating surface (the adhesive is not very viscous, so it will spread and spill if over applied).



17. Install the tool again to the same three points of the cylinder head as previous. Again, the base plate retaining bolts should be torqued to approximately 13 +/- 3Nm to ensure the base plate is sufficiently seated.

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18. Turn the installation screw clockwise until the outer flange of the Core Plug installation head contacts firmly the flat, machined section of the cylinder head (the installation screw will become hard to turn). **Caution:** Do not turn the installation screw past the contact point between this flange and the cylinder head or damage may occur to the tool or cylinder head.



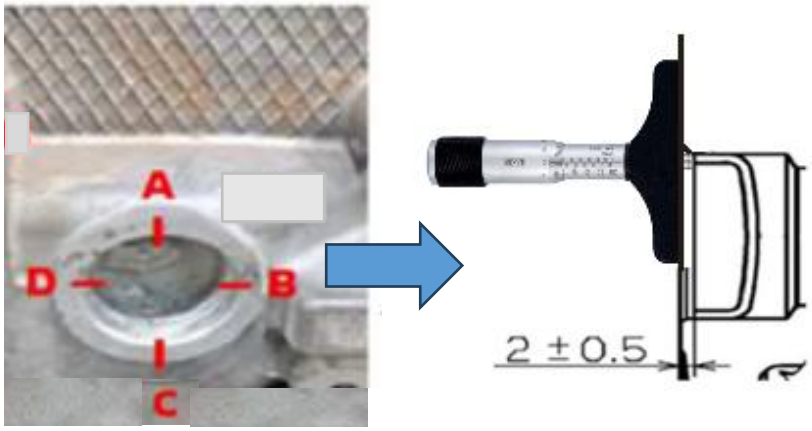
19. Turn the installation screw anticlockwise to release it from the core plug then remove the tool assembly.

20. Clean off any residual adhesive with brake cleaner (or similar) and a lint free cloth. Inpsect the installation condition of the core plug and measure the installation depth at points A, B, C & D using a depth micrometre or similar.



Depth from the outer lip of the core plug to the machine surface of the head should be 2mm +/- 0.5mm (with the maximum allowable difference between any two points being 0.5mm). Record these measurements on your repair order and warranty claim.

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If the measurements are outside of specification, inspect the cylinder head and installation tool for damage. If no damage is noted, install a new core plug following the removal and installation steps [8] to [20] again.

21. Reinstall all components removed in steps [2] to [7] in reverse order, ensuring to align the hood using the tape markings made previously and remove protective tape/hose/etc during the process.

22. Refill, bleed and pressure test the cooling system. If leakage is present at the core plug directly after repair and measurements taken at step [20] were within specification, replace the cylinder head as per workshop manual instructions.

End Procedure

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